# Vroomie
# Audio Diagnostics Architecture & Engineering Handover

Complete technical reference for architects and development teams extending the Vroomie audio anomaly-detection system

Engine version v9.7 · ETHANOL-FEATURE
github.com/DGeorgeA/vroomie-app · branch main
Live at vroomie.in

All performance figures herein are measured results on held-out data, reproducible with the committed test harnesses.

# 1. The one-page mental model

Vroomie now runs **two matchers with opposite strengths over one capture pipeline**:

```
Microphone (AGC/NS/EC requested off; actual settings recorded)
  → native-rate capture → 1 s ring buffer (~0.9 s cadence) → resample 16 kHz
        │
        ├─► TIER 1 — CONSTELLATION FINGERPRINT (Shazam-style)  [≤5 s, exact recordings]
        │     rolling 5 s window → spectral-peak hashes → time-offset coherence
        │     HIT   identify + auto-finalise session (97%, matchMethod:
        │           acoustic_fingerprint). No hit   completely silent.
        │
        └─► TIER 2 — EMBEDDING ENGINE (YAMNet)  [generalising, session-based]
              silence gate → RMS-normalize 0.05 → YAMNet → domain gate
              → cosine vs 352 fault embeddings + 94 anchors → margin rule
              → session vote (primary fraction / recovery vote)
              → possibility 70–97% statement
```

| | Tier 1: Constellation | Tier 2: Embeddings |
|---|---|---|
| Question answered | "Is this **exact recording** playing?" | "Does this sound **like** a known fault family?" |
| Latency | ~3–5 s, auto-finalises | full session (user stops) |
| Generalisation | none — by design | the whole point |
| Bench replay of a Supabase sample | **primary owner** | fallback |
| Real customer vehicle | blind (no shared hashes) | **primary owner** |
| Decision statistic | time-offset coherent votes | cosine margin over anchors |
| False-positive defence | coherence itself (unrelated audio can't align) | anchors + domain gate + session vote |

The tiers are **additive**: Tier 1 never suppresses Tier 2; an unavailable fingerprint index simply means Tier 2 runs alone (fail-safe measured, not assumed).

# 2. Why a percentage threshold cannot live in Tier 1

The product question "if a squeak or bearing on a *different car* sounds similar, shouldn't ~65–72% match categorize it?" decomposes onto the two tiers:

- **Constellation scores are not similarity percentages.** A coherent score of 1400 vs 237 is not "6× more similar" — it counts hash pairs that agree at one time offset. Two *different recordings* of the same fault share almost **zero** hashes: this is the same property that makes Shazam identify a studio track but fail on a cover version. There is no threshold at which Tier 1 recognises a different vehicle — 65%, 72% or any other number. Lowering its thresholds does not add generalisation; it only admits random coherence (noise).
- **The 65–72% band already has a home: Tier 2's margin scale.** The published possibility is `70 + 108×(margin − 0.04)` clamped to [70, 97] — so "65–72%" corresponds to margins just under the 0.04 confirm threshold. That is where a "possible match" tier belongs, and where it was measured (§6).

**Rule for future tuning: route exact-replay demands to Tier 1, similarity demands to Tier 2. Never tune one tier to do the other's job** — each failure mode of the past (hallucination, threshold-loosening, 20% healthy FP) came from asking a similarity system to behave like an identifier or vice versa.

# 3. Tier 1 in detail (what shipped, and the review that changed it)

**Algorithm** (`src/lib/constellationMatcher.js`, self-contained, no imports): 1024-pt FFT, 16 ms hop; per-band strongest bin above 1.6× band mean (6 log-spaced bands); anchor–target pairing (fan-out 6, Δt ≤ 48 frames) hashed as `(f1, Δf, Δt)`; matching by per-reference **offset histogram** — the score is the tallest single-offset bin. The index builder **imports the same hashing function**, so index/query parity is structural, not conventional.

**Index** (`public/constellation_v1.json`): 9 references (6 originals extended to 10 s + intake/misfire/timing at 5 s), 166 k entries, 1.7 MB, service-worker cached. The 10 s extensions use **equal-power crossfade looping** (`scripts/extend_reference_wavs.py`) because naive concatenation produces click transients that become spurious spectral peaks.

**Runtime behaviour**: rolling 5 s listener fed un-normalized 16 kHz audio (peaks are relative maxima — already level-invariant); earliest fire at 3 s; a hit toasts "Identified: X — matched in Ns", auto-finalises the session, bypasses abort gates (the audio *was* identified), and stores `matchMethod: 'acoustic_fingerprint'` with the coherence score.

## 3.1 Pre-ship review — flaws found and fixed

| # | Flaw (measured) | Fix (measured) |
|---|---|---|
| F1 | **Threshold provenance**: thresholds derived on a 6-ref prototype; against the *shipped* 13-ref index the worst healthy negative scored **443 — above the 400 raw floor** (was 237). | Review protocol re-measures through the shipped hydrate/match code path. Floor raised **400** → **600**; weakest passing positive condition is 621, so nothing that worked stops working; headroom restored to ≥1.4× on both criteria. |
| F2 | **Short-reference bias**: normalizing by query hashes alone means a 1.5 s reference physically cannot reach 0.05 with a 5 s listen — a genuine replay of an indexed 1.5 s file scored 385/0.035: **failed against itself**. | Normalization corrected to `score / min(queryHashes, refHashes)`; per-ref `hash_count` now ships in the artifact. Additionally a measured **4 s minimum reference length** — the four 1.5 s PS variants were dropped (their replay detection falls to Tier 2, which measurably handles them). |
| F3 | **Single-play vs looped**: does one playback of a 5 s file inside the listen window still match? | Measured: intake 1574/0.144, misfire 676/0.062, timing 766/0.071 — all pass identically single-played or looped. |
| F4 | **Ambiguity from near-duplicate references** | Measured non-issue: second-best scores 65–118 vs best 1388–2516. |

## 3.2 Tier 1 operating point (current, all re-measured on the shipped index)

```
MIN_COHERENT_SCORE   = 600      worst negative 443  → 1.4× headroom
MIN_NORMALIZED_SCORE = 0.05     worst negative 0.040 → 1.25× headroom
                                weakest passing positive: 621 / 0.06
                                strongest negative set: 115 healthy clips,
                                4 real speech recordings (direct + speaker-
                                colorized), music, fan, traffic
LISTEN_SECONDS = 5, earliest fire 3 s
```

**In-app proof** (served build, speaker-colorized audio injected as microphone): Piston → "Identified: Piston" at 3 s, auto-finalised, 97%; speech → no hit, correct diagnostic abort. Post-recalibration playback modes of the 2 s original: looped 1211/0.109, single-play 947/0.213, extended-10 s excerpt 4952/0.449 — all comfortably above the 600/0.05 thresholds.

## 4. Tier 2 in detail (the generalising engine, v9.9)

Unchanged by the constellation work — summarised for completeness; full history in `VROOMIE_ARCHITECTURE_HANDOVER.md`.

- **References**: offline factory (`build_reference_fingerprints.mjs`) → 352 embeddings, per-family cap 100 (power_steering was 58.8% of the set; the cap measurably cut healthy FPs 9→6/140 and fixed two speaker-coloration misclassifications), 8 augmentation variants including the harsh phone-speaker channel; 94 anchors (healthy + interferers incl. traffic).
- **Gate**: vehicle top-1 | weak generic | vehicle-evidence | **speaker-rescue** (interferer top-1 with vehicle evidence ≥0.02 and interferer ≤0.30 — measured 29/70 vetoed fault windows recovered, 0/108 speech/music admitted).
- **Match**: cosine ≥0.45 **and** margin over best anchor ≥0.04.
- **Session**: primary fraction rule (≥45% of accepted windows, one family, ≥3 windows) + v9.8 recovery vote (≥60% total candidates → plurality; similarity-sum tie-break at 1.10 dominance, else decline).
- **Diagnosability**: every abort persists a hidden `status:'rejected'` row with per-stage counts, what YAMNet heard, and the device's applied capture settings.

## 5. Implications of constellation matching for application finetuning

1. **Bench-testing is no longer a calibration signal for Tier 2.** Replayed Supabase samples will be captured by Tier 1 almost immediately. Any future "the sample isn't detected" report first asks: did Tier 1 miss (index/threshold issue — check `matchMethod` and coherence scores) or did it correctly fall through (capture too degraded — the diagnostics row tells you)? **Do not retune Tier 2 thresholds from bench replays anymore** — they will rarely even reach it.
2. **The reference library's two roles have diverged.** For Tier 1 a reference is a *recording identity* — longer is strictly better (more frames → taller coherence spike), and near-duplicates add nothing. For Tier 2 a reference is a *class sample* — diversity matters, duplicates skew the prior (the 58.8% dominance problem). Curate accordingly: one long canonical recording per identity for Tier 1; many *distinct* real recordings per family for Tier 2.
3. **Adding a new fault class** now takes three steps, all automated after upload: put a ≥4 s (ideally ≥10 s) real recording in the bucket → `node scripts/build_constellation_index.mjs` → `node scripts/build_reference_fingerprints.mjs`; then re-run `review_constellation_shipped.mjs` and the QA suites. The 4 s index minimum is enforced automatically; sub-4 s files still serve Tier 2.
4. **Index growth needs threshold re-validation.** F1 is a standing lesson: negative scores grow with index size (more chances of random alignment). The review script *is* the gate — re-run it whenever `reference_count` changes, and treat its worst-negative line as the floor-setter.
5. **Latency identity.** Tier 1 gives the product its "Shazam moment" (~3 s). Nothing in Tier 2 should be tuned to chase that number — its value is generalisation, and its honest latency is a session. The UI already reflects this split (instant "Identified" toast vs end-of-session report).
6. **What still cannot be finetuned into existence:** recognising a *different* vehicle's similar-sounding fault at high confidence. That requires data (real phone-mic recordings of real faulty vehicles feeding Tier 2), not

thresholds. This remains the single highest-leverage investment in the product.

# 6. The 65–72% "possible match" tier (Tier 2 extension)

**Semantics**: a window is a *near* candidate when similarity ≥0.45 and margin □ [0.02, 0.04) — i.e. the sound is much closer to a fault family than to any healthy/noise anchor, but below the confirmed band. A session reports **"possible □family□ — verification required"** at 65–69% only when nothing was confirmed and near+full candidates of one family reach the same 45% fraction.

**Measured feasibility** (scripts/measure_near_match_tier.mjs, 140 held-out healthy clips + known acoustic misses + interferer suite):

| Axis | Result |
|---|---|
| **Value** — known acoustic misses recovered | **0 of 3.** Rocker@2m and MotorStarter now confirm outright under v9.9 (the tier is moot for them); the alternator-critical replay dies at the domain gate, so no margin band can reach it. |
| **Guard** — interferers entering the tier | 0 (fan, traffic, white, music, 4 real speech recordings — all clean) |
| **Cost** — additional healthy "possible" alarms | **+8 of 140 (5.7%)** — on top of the 6 confirmed FPs, healthy vehicles seeing an alarm would rise from 6 to 14 (4.3% → 10%) |

## 6.1 Decision: NOT SHIPPED — rejected on measurement

The tier produced **zero measured recall benefit and a near-doubling of healthy-vehicle alarms**. Eight healthy engines would have been told "possible power steering / rocker / bearing issue" for nothing. The intuition behind the request is still served, in the two places it belongs:

- **Same recording, degraded capture** → Tier 1 already matches far below "72% similarity" conditions (harsh speaker, 50 cm, noise) because coherence, not similarity, is the criterion.
- **Different vehicle, similar fault** → this is a *data* problem. The 65–72% embedding band is dominated by healthy engines on the current reference library; the band becomes usable exactly when the library contains diverse real recordings per family (see §5.6). Re-run `measure_near_match_tier.mjs` after each significant library expansion — the day its cost line drops near zero, the tier can ship with the same code that measured it.

# 7. Operating limits (honest)

- Tier 1 identifies **these exact recordings**. It will never recognise a different car's fault; that is Tier 2's job, and Tier 2's recall is data-bound (~72% held-out).
- All acoustic figures derive from **channel simulations**; the definitive speaker→air→microphone test runs on a physical device, and every session (including failures) now leaves remotely readable telemetry for exactly that purpose.
- One device-class risk stands open: OS-forced noise suppression, now visible per-session in `capture_settings`.

# 8. Harness map (run these, don't rebuild them)

| Question | Harness |
|---|---|
| Tier 1 thresholds still safe? | `review_constellation_shipped.mjs` (re-run after ANY index change) |
| Tier 1 sensitivity across capture conditions | `proto_constellation_sensitivity.mjs` |
| Tier 2 operating point / FP baseline | `rca_healthy_fp_sweep.mjs` (all 140 — never a stride sample) |
| Gate behaviour under speaker coloration | `rca_gate_boundary.mjs`, `rca_gate_fix_validation.mjs` |
| Session-rule calibration | `rca_tiebreak_probe.mjs`, `rule_explorer.mjs` |
| End-to-end in the real app | fake-mic injection E2E (see handover doc §7.1) |
| Fast unit/policy assertions | `qa_unit_tests.mjs`, `qa_ethanol_feature.mjs` |